

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of CSE

Program: B.Sc. in CSE

Mid Term Examination, Autumn 2025

Course Title: Electronics Device and Circuits Lab

Course Code: EEE 0714211/EEE 251

Marks: 20

Time: 1(one) hour

(Answer all questions)

Q.No.

Questions

Marks

1. a) Briefly explain the I-V characteristics of p-n junction diode in forward and reverse bias condition with necessary diagrams. Also write the differences between n type and p type material.

[05]

- b) Find the output voltage V_O of the circuit shown in figure 1b with necessary explanation and sketch the output voltage.

[05]

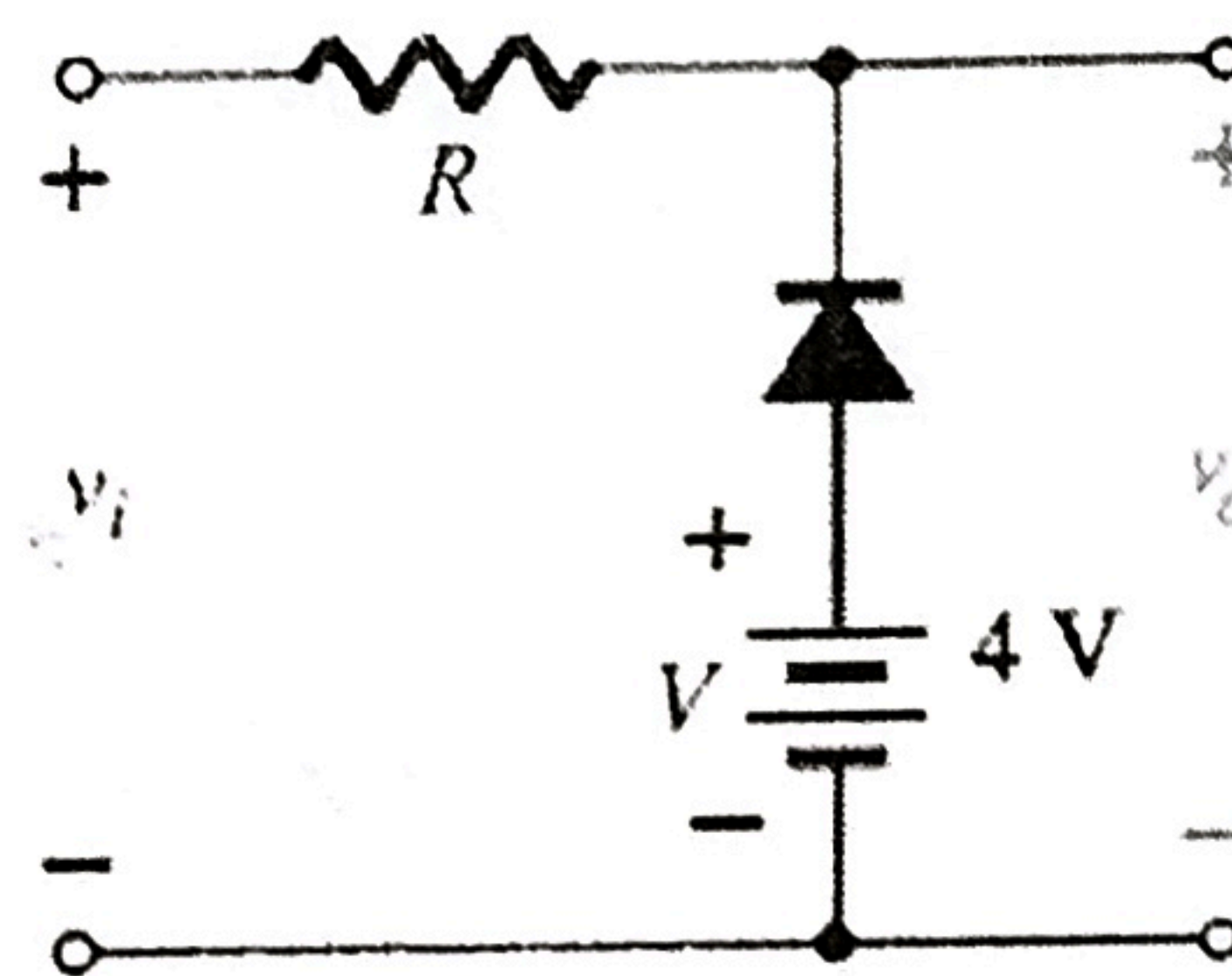
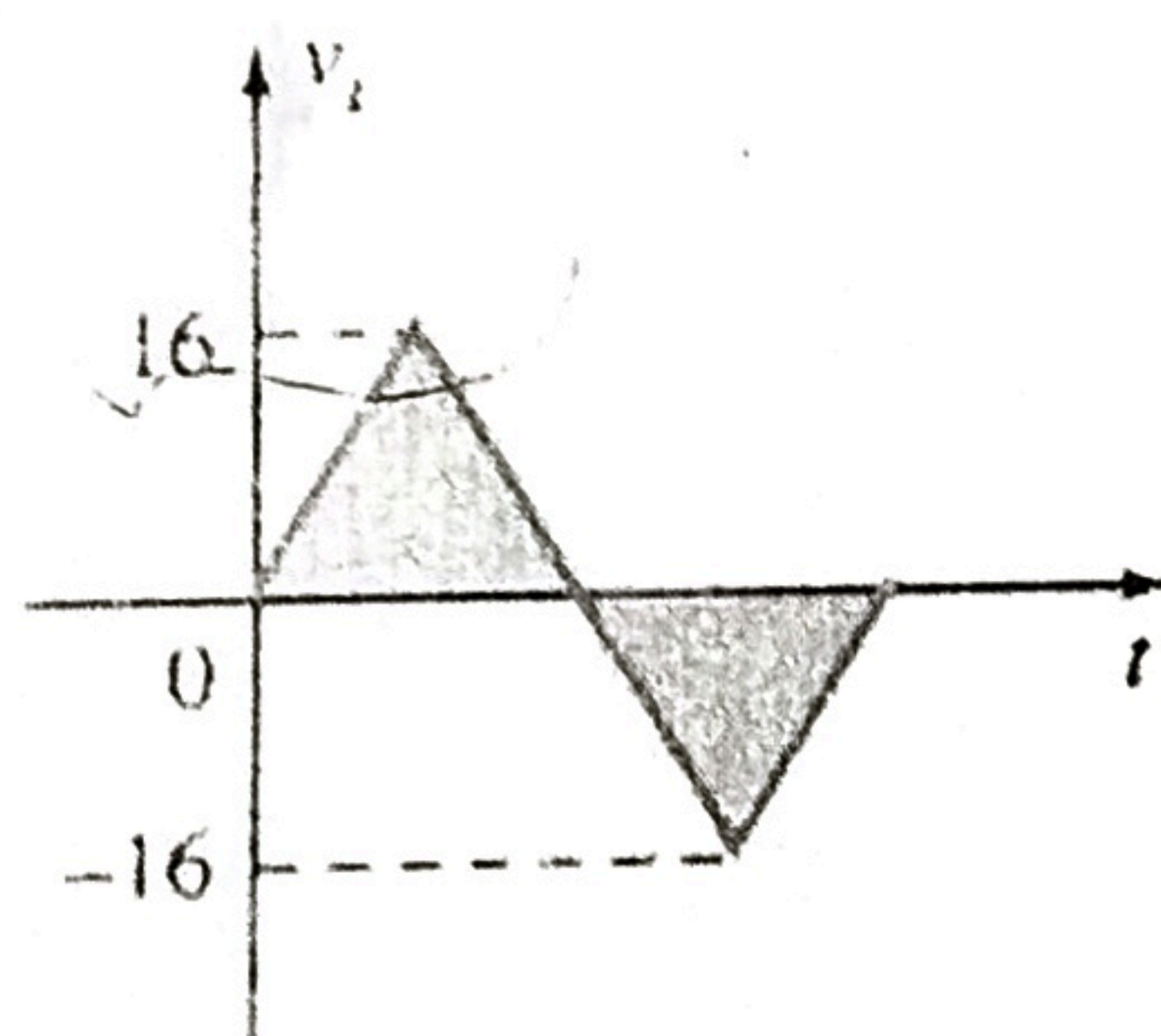
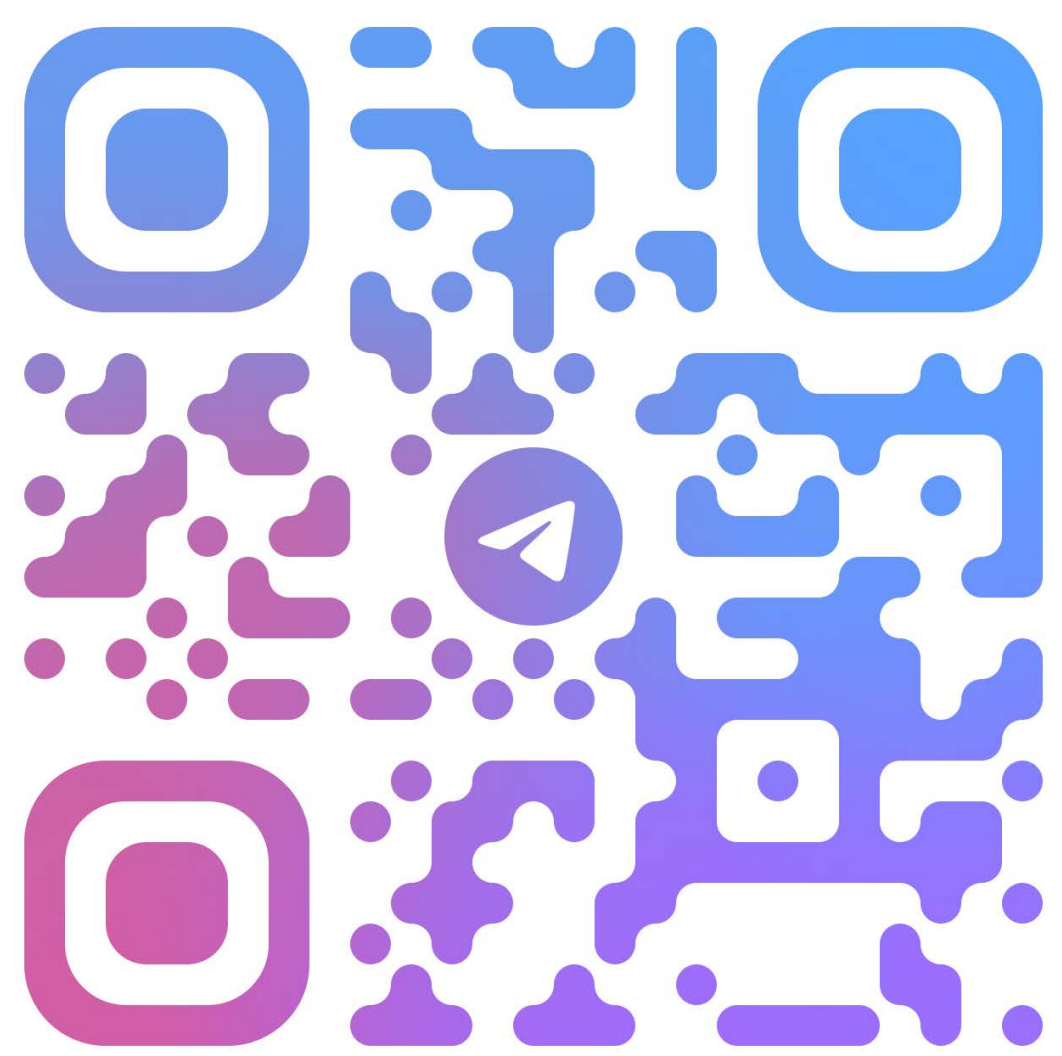


Figure 1(b)

2. a) Describe the operating principle of half wave rectifier and full wave rectifier circuit.

[05]



- b) Determine the output voltage V_o of the circuit shown in figure 2b with necessary explanation and sketch the output voltage. Here $V_1 = 5V$ [05]

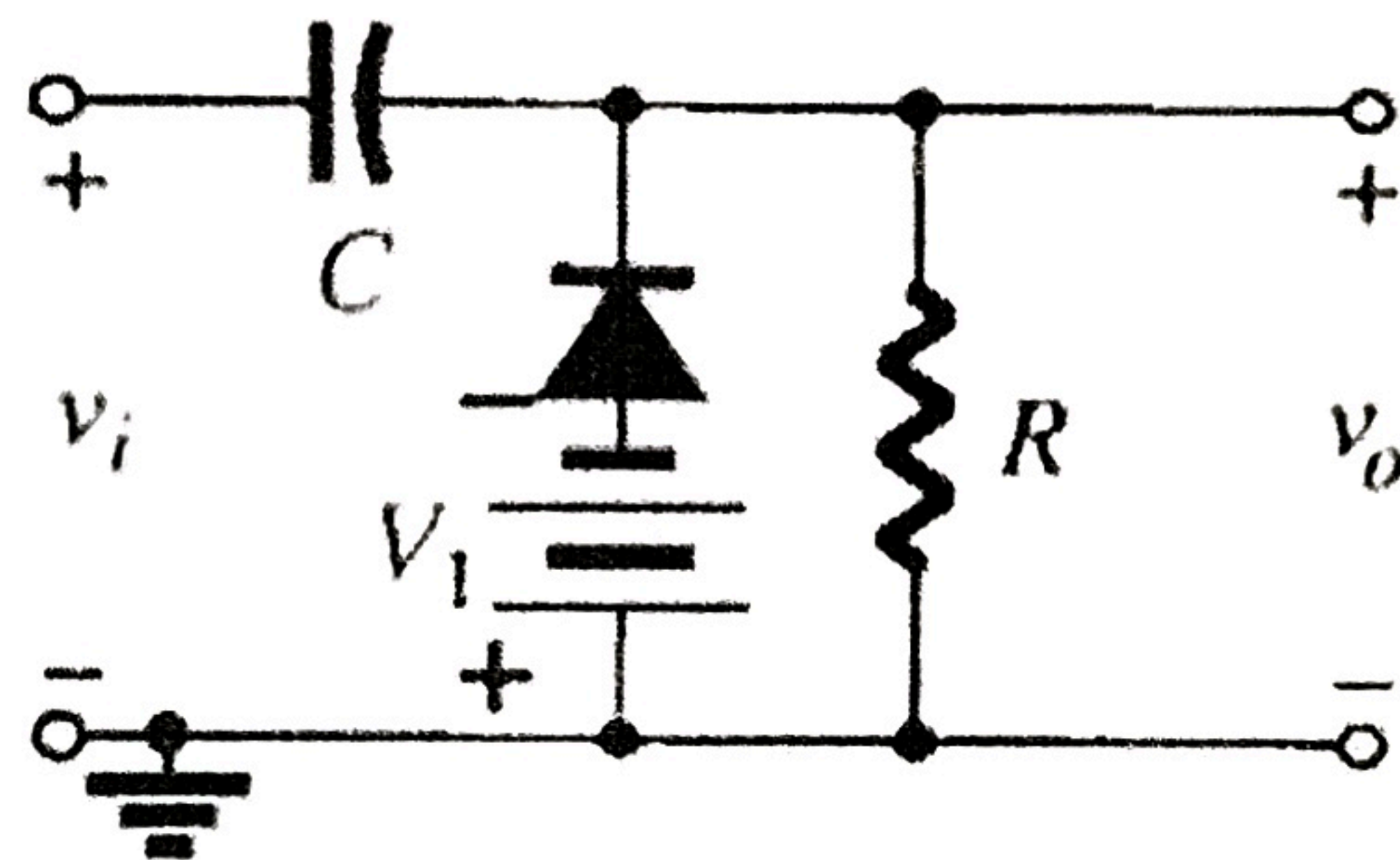
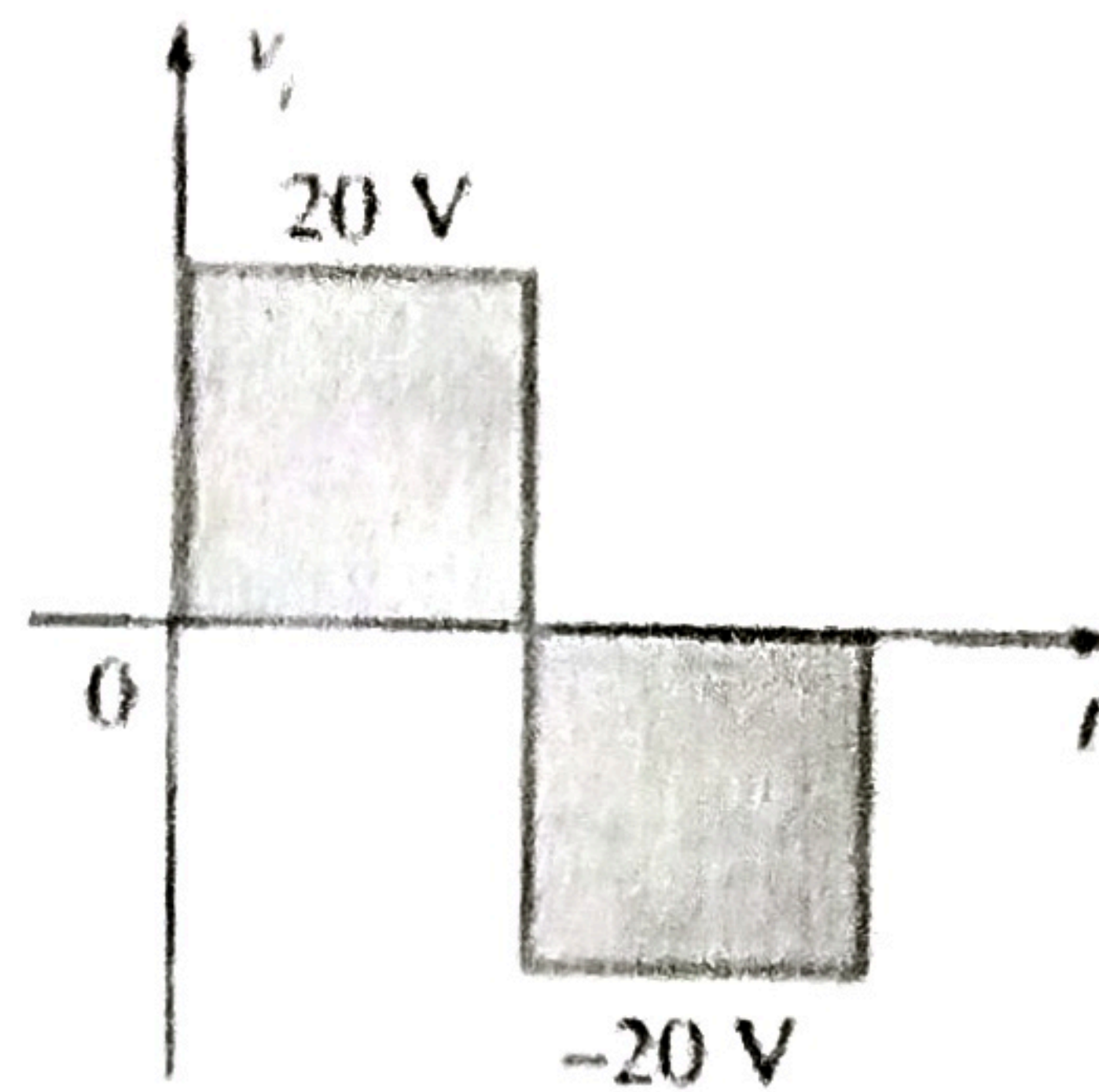


Figure 2(b)

